

Research defense and next questions

Project: Reproducible AI Research Study

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Explain the mechanism and terminology of research defense and next questions

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Presentation and challenge session

Without opening the old lesson, explain one key idea from Presentation and challenge session. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of presentation and challenge session

Reference example, not the only acceptable wording: Teacher explanation: A presentation should make the evidence easy to challenge. Show the baseline, scope, negative cases and uncertainties instead of only a best example. Use a short sequence: question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating evidence or stating it is missing.

Independent reproduction check

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of independent reproduction check

Reference example, not the only acceptable wording: Teacher explanation: Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help. Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

Writing the methods section

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of writing the methods section

Reference example, not the only acceptable wording: Teacher explanation: A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect evidence. Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

Two 45-minute sessions. A research defense distinguishes what was learned, what failed and which next question is justified. A complete investigation can end with a limited or negative finding. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the mechanism and terminology of research defense and next questions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

End goal for the pictorial case

Explain research defense and next questions using the records and boundaries of this fictional investigation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

A research defense distinguishes what was learned, what failed and which next question is justified. A complete investigation can end with a limited or negative finding.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term · Meaning

Term: Research defense

Meaning: A justified account of methods, findings and limits

Term: Next question

Meaning: A new bounded inquiry motivated by a gap

Term: Negative finding

Meaning: A result failing the declared expectation

Research defense and next questions — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study.

Check or calculation: Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.

Conclusion and limit: 75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested. Qualification: The next question follows evidence gaps; it must not be presented as an answer already obtained. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study.

75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Research defense and next questions

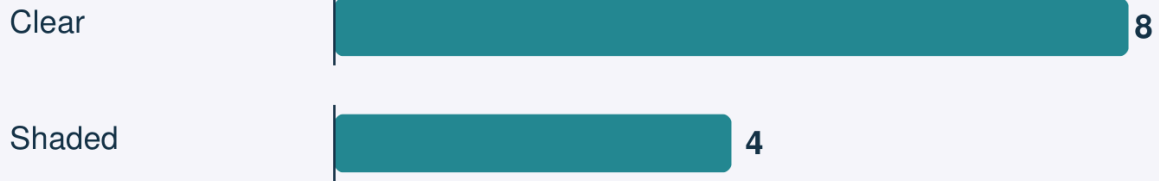
Defend the next research question. A research defence identifies a observed weakness and a testable follow-up.

The facts we will use

A frozen model is correct on 8/10 clear images and 4/10 shaded images. Source objects are grouped within each condition.

Defence: different condition slices

correct cases out of 10 per slice



Bar length shows the supplied value; the marked baseline is zero.

Research defense and next questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State slice outcomes

Clear is 80%; shaded is 40%.



2. Limit the interpretation

Shade sensitivity is plausible, but camera or object differences may also matter.



3. Design follow-up

Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

CHANGE ONE THING

Change cameras as well as lighting in the follow-up.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. State slice outcomes

What happens and why: Clear is 80%; shaded is 40%.

Step: 2. Limit the interpretation

What happens and why: Shade sensitivity is plausible, but camera or object differences may also matter.

Step: 3. Design follow-up

What happens and why: Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

What this example does not establish

Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

Change one thing in the pictured case

Change: Change cameras as well as lighting in the follow-up.

Prediction: The design again entangles two conditions.

Reason: A camera difference provides another explanation for any performance gap.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Time or stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Time or stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Time or stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Time or stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Time or stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Time or stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Time or stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Reduce help deliberately

Completion task: Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before

discussion.

Individual reasoning check

Use the worked case for Research defense and next questions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The next question follows evidence gaps; it must not be presented as an answer already obtained.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

Reasoned teacher answer

Clear is 80%; shaded is 40%. Shade sensitivity is plausible, but camera or object differences may also matter. Photograph the same held-out objects under both light conditions using one camera and fixed scoring. Changed case: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Changed-case reasoning: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

The worked result establishes every future case: The next question follows evidence gaps; it must not be presented as an answer already obtained.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: 75% versus 68.75%, gain 6.25 points in 16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested. Qualification: The next question follows evidence gaps; it must not be presented as an answer already obtained. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

Teacher explanation: A research defense distinguishes what was learned, what failed and which next question is justified. A complete investigation can end with a limited or negative finding. Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.

W2 Trace the worked case

Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

W3 Audit the limit

A report omits this qualification: The next question follows evidence gaps; it must not be presented as an answer already obtained. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: The next question follows evidence gaps; it must not be presented as an answer already obtained. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Project candidate9/12, baseline10/12; reproduction matches. State result and a useful next question.

Teacher explanation: 75% versus83.333%, candidate lower8.333 points. Investigate recorded failure conditions with a revised prespecified method and fresh reserved cases; do not erase the negative result.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. The next question follows evidence gaps; it must not be presented as an answer already obtained. The new case should test this particular mechanism rather than merely restate the original result.

Independent assessment and fresh retry

W4: 75% versus83.333%, candidate lower8.333 points. Investigate recorded failure conditions with a revised prespecified method and fresh reserved cases; do not erase the negative result. Require intermediate reasoning and the stated scope. Fresh retry: Both procedures8/10 and successful replay.

Expected reasoning: Both 80% in this sample; strict improvement not observed. Equivalence is not established without its own design. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case: Project candidate 9/12, baseline 10/12; reproduction matches. State result and a useful next question.

Selected case answer

Teacher explanation: 75% versus 83.333%, candidate lower 8.333 points. Investigate recorded failure conditions with a revised prespecified method and fresh reserved cases; do not erase the negative result.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final

evidence pack: Project candidate9/12, baseline10/12; reproduction matches. State result and a useful next question. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Investigate a bounded question with justified baselines, controlled experiments, uncertainty and reproducible evidence, then defend the findings and their limitations.. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

7. Prepare, teach and assess

Teaching target: Explain the mechanism and terminology of research defense and next questions

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Check every learner before discussion: Use the worked case for Research defense and next questions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The next question follows evidence gaps; it must not be presented as an answer already obtained.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change cameras as well as lighting in the follow-up. Require a prediction and a reason before comparing. Expected result: The design again entangles two conditions. Reason: A camera difference provides another explanation for any performance gap.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.